

WGCNA Analysis of lncRNA Expression Data: Pipeline Structure, Results, and Signed vs. Unsigned Network Comparison

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1 Pipeline Structure

The lncRNA script (`lncRNA_WGCNA.R`) follows the exact same stage order as the mRNA pipeline, so this section only covers the parameters and logic that differ from the mRNA run.

1.1 Global Parameters

- `MIN_COUNT = NA`, `MIN_FRAC = NA`. The low-expression filter was disabled for this run. The supplied file (`lncRNA_filter1.xlsx`) had already been expression-filtered upstream. All 4115 lncRNA genes were therefore carried into normalisation and network construction.
- `N_GENES_WGCNA = NA`, `MAX_BLOCK_SIZE = 12000`, `N_HUBS = 20`.

1.2 Loading and Size-Factor Reuse

The file (`lncRNA_filter1.xlsx`) is the same 963-sample cohort with 4115 lncRNA count columns plus Subtype. As with the mRNA script, DESeq2 is run with an intercept-only design (~ 1) purely as a normalisation/VST tool. Critically, **size factors are not re-estimated here**: they are *imported directly from the mRNA run's saved `mrna_size_factors.rds`, matched to the lncRNA cohort by sample name (with an explicit check that all 963 samples have a match before proceeding)*. The console confirms that all 963 of 963 samples matched, and the imported size factors span 0.374 to 2.048 (median 1.029), identical to the mRNA run as expected.

Justification: the lncRNA count matrix has a substantially higher zero fraction than the mRNA matrix (17.2% vs. 5.1%), and library depth is a property of the sequencing run rather than of transcript biotype, so the size factors estimated from the denser mRNA counts are the more reliable per-sample depth correction for both data types.

2 Results

2.1 Cohort and Preprocessing Summary

The cohort is identical to the mRNA run: 963 samples, PAM50 subtype counts Basal 170, HER2 75, LumA 528, LumB 190. **All 4115 lncRNA genes in the supplied matrix were carried forward, as no additional low-expression filter was applied. The matrix contains 17.2% zero entries, against 5.1% for mRNA.** Per-sample lncRNA library totals (median 250,574; range 62,624–625,909) are roughly two orders of magnitude smaller than the mRNA library totals. **All 963 samples and all 4115 genes entered network construction**.

2.2 MAD Distribution

Figure 1 shows the same right-skewed shape seen in the mRNA MAD histogram: most of the **4115** genes cluster at low-to-moderate MAD (roughly 1.0–1.5), with a long thin tail of more variable genes extending out past $\text{MAD} = 4\text{--}6$.

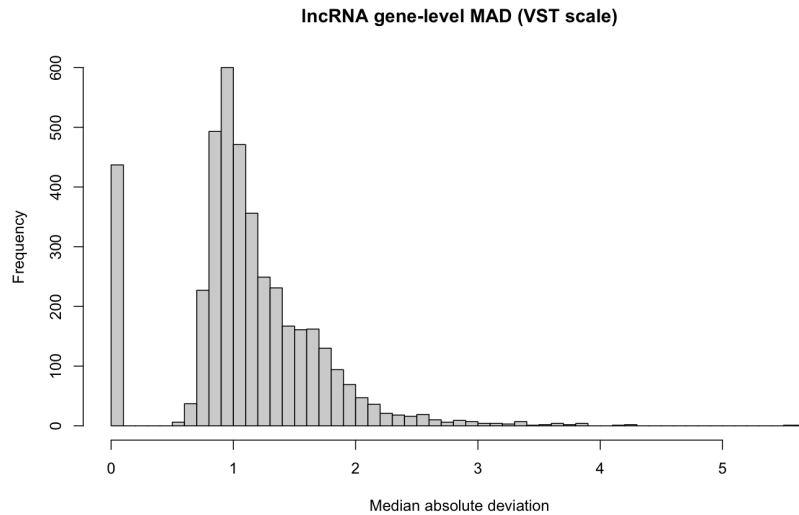


Figure 1: Distribution of gene-level MAD (VST scale) across the **4115** lncRNA genes.

2.3 Sample Clustering

Figure 2 shows the same style of outlier-screening dendrogram used for mRNA. No sample or small cluster of samples sits at a dramatically greater height than the rest of the tree, so no cut height was applied and all 963 samples were retained.

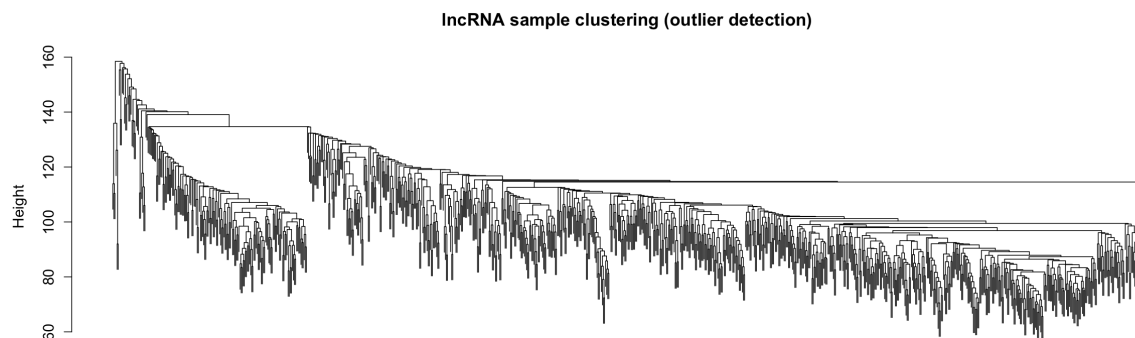


Figure 2: Average-linkage hierarchical clustering of all 963 samples on lncRNA expression (outlier screening). No cut was applied.

2.4 Soft-Threshold Selection

Figure 3 and Table 1 summarise power selection. The signed network's fit curve rises smoothly and monotonically, crossing $R^2 = 0.90$ at power **12** ($R^2 = \mathbf{0.907}$). The **unsigned** fit curve, by contrast, is unstable at higher powers: it rises smoothly to $R^2 = 0.961$ at power 7 and stays above 0.92 through power 10, then collapses to $R^2 = 0.366$ at power 12 and remains below 0.43 at every power from 12 to 20.

Table 1: Selected soft-thresholding power and fit statistics for each lncRNA network type.
Values updated for the unfiltered run.

Network type	Chosen power	Scale-free R^2 at chosen power	Mean connectivity
Unsigned	5	0.911	1.85
Signed	12	0.907	4.37

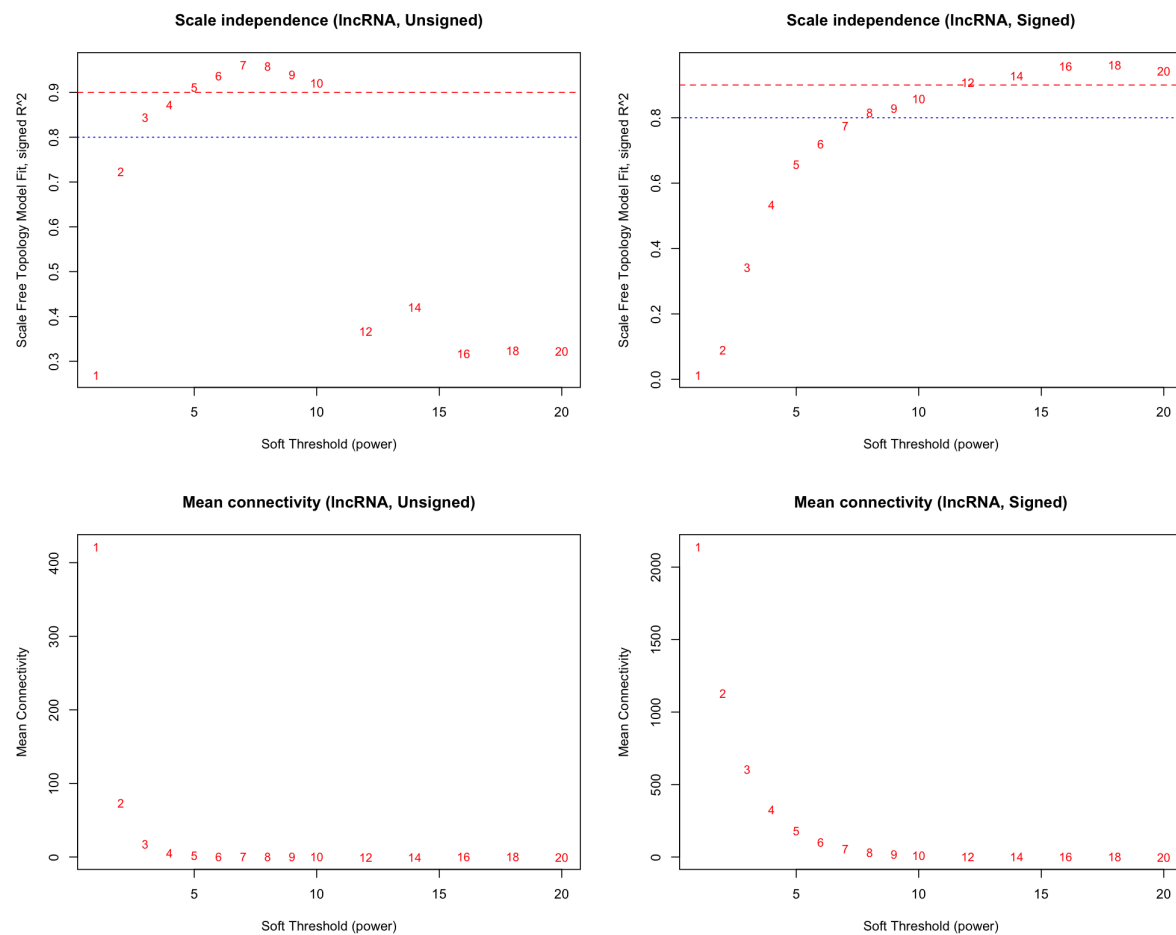


Figure 3: Scale-free topology fit and mean connectivity vs. soft-thresholding power for the unsigned (left) and signed (right) lncRNA networks. Note the unsigned fit's instability above power **10**.

2.5 Network Construction and Module Detection

Table 2: lncRNA module sizes under both parameter sets.

Module	Signed		Unsigned	
	First run (used)	First run (used)		
black	91	89		
blue	329	218		
brown	327	217		
green	131	128		
magenta	58	47		
pink	81	80		
purple	38	—	—	
red	113	106		
turquoise	362	349		
yellow	141	140		
grey (unassigned)	2444	2741		
Total non-grey	1671 (40.6%)	1374 (33.4%)		
Total genes	4115	4115	4115	4115

The **signed** network organises a larger number of lncRNA genes into non-grey modules than the unsigned network (40.6% vs. 33.4%), which is the same direction as the mRNA result and the reverse of what was observed in the previous, expression-filtered lncRNA run. It also resolves ten non-grey modules against nine for unsigned, the extra one being **purple** (38 genes).

Figure 5 shows the gene dendrograms for both network types under the first-pass parameters chosen for this analysis.

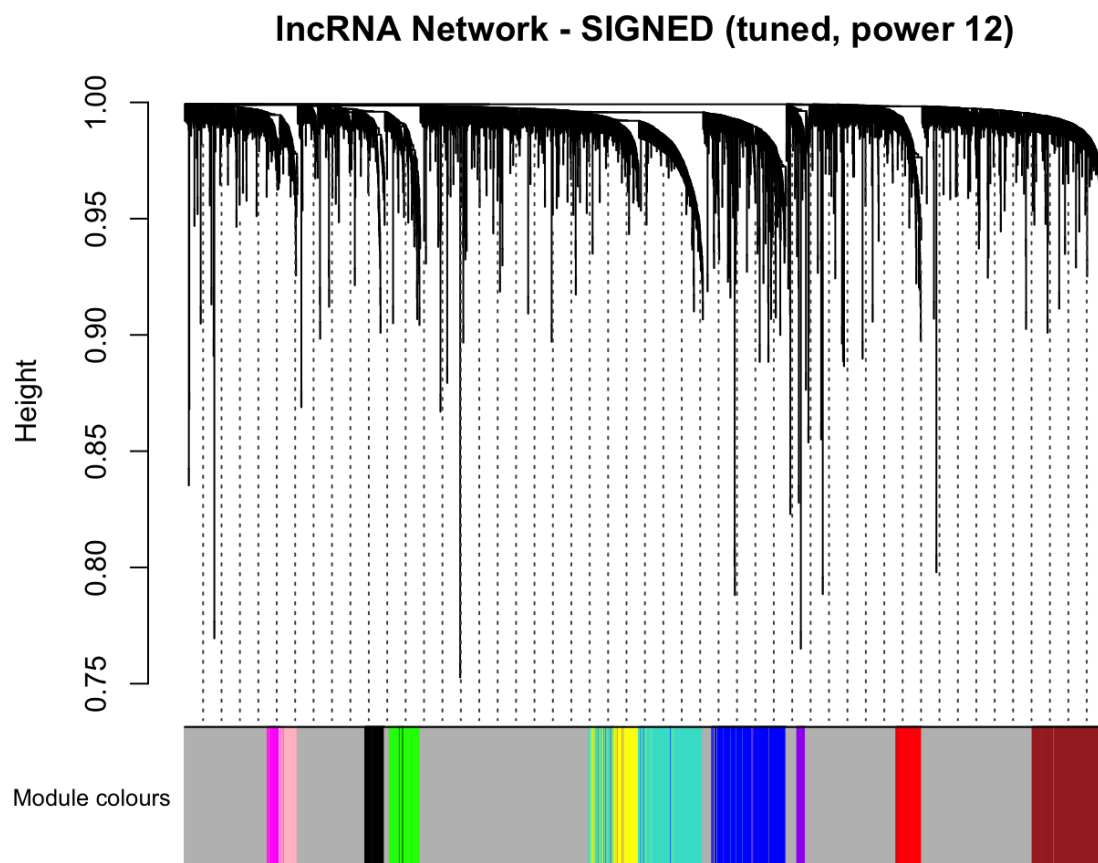


Figure 4: Gene clustering dendrogram with module colours, signed lncRNA network (power **12**, first-pass parameters). Module colour bands are largely solid, contiguous blocks corresponding to single well-separated branches.

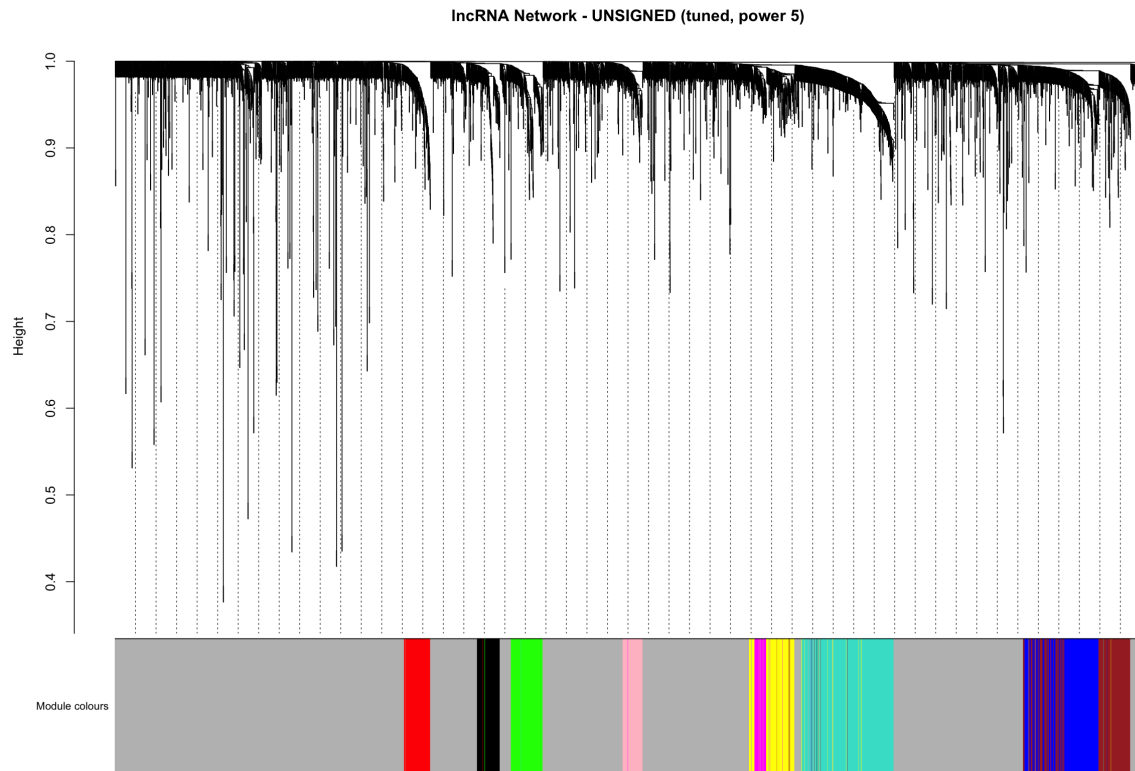


Figure 5: Gene clustering dendrogram with module colours, unsigned lncRNA network (power 5, first-pass parameters). Several module colours (e.g. yellow, blue, turquoise) appear as multiple separated stripes rather than one contiguous block.

2.6 Module Eigengene Correlation Structure

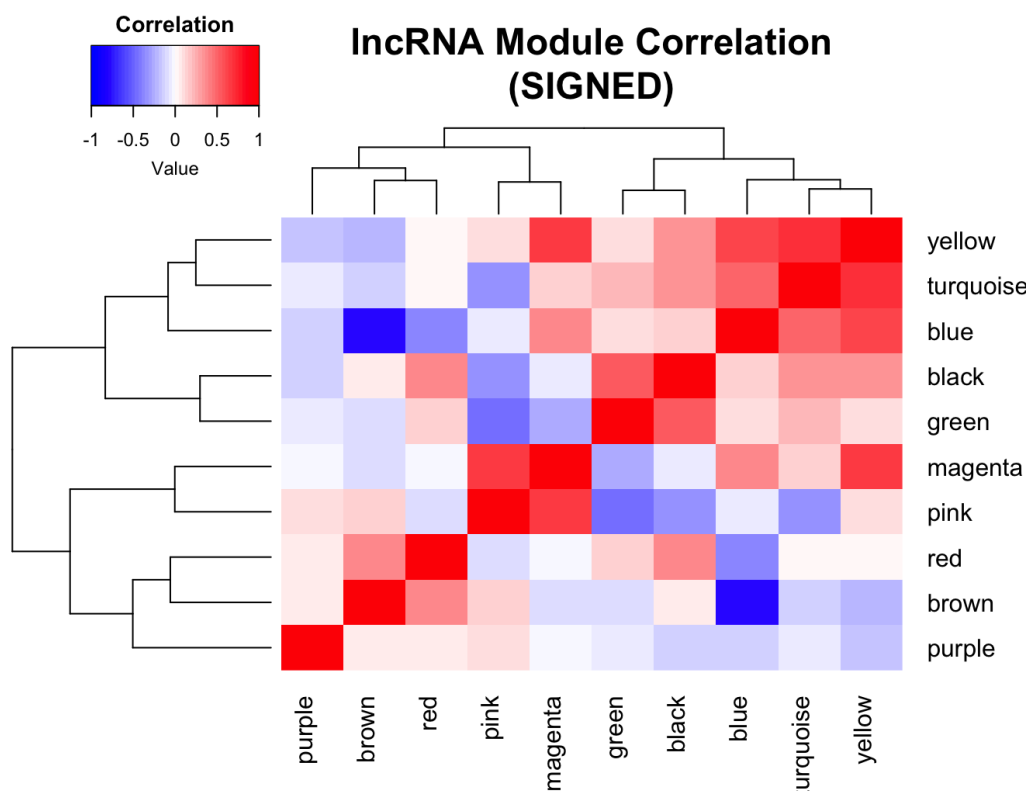


Figure 6: Module eigengene correlation heatmap, signed lncRNA network. **Figure file updated.**

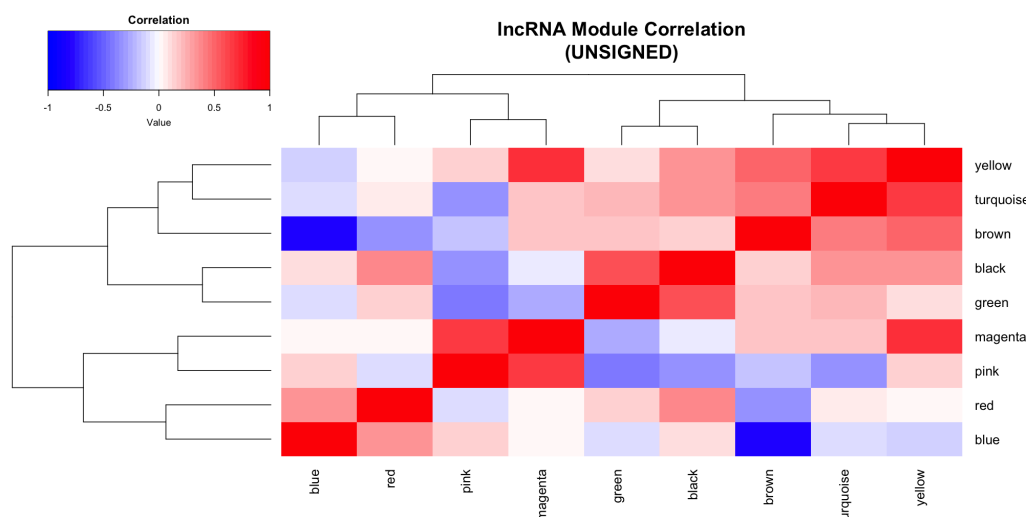


Figure 7: Module eigengene correlation heatmap, unsigned lncRNA network. **Figure file updated.**

2.7 Module–Trait Relationships

Table 3 and Figure 8 report the signed module–subtype correlations; Figure 9 reports the unsigned equivalent. As with mRNA, the dominant pattern is a Basal/LumA opposition, led here by **blue** ($r = -0.81$ Basal, $r = 0.59$ LumA) and **brown** ($r = 0.89$ Basal). *Unlike the*

mRNA result, HER2 shows more modules with a detectable association here (e.g. yellow $r = -0.22$, blue $r = -0.21$) , and LumB shows a clearer distinguishing signal in a subset of modules (black $r = -0.30$, brown $r = -0.31$) .

Table 3: Signed module–subtype Pearson correlations, lncRNA network.

Module	Basal	HER2	LumA	LumB
brown	0.89	−0.05 (n.s.)	−0.41	−0.31
blue	−0.81	−0.21	0.59	0.18
yellow	−0.35	−0.22	0.41	−0.03 (n.s.)
red	0.26	0.10	−0.15	−0.13
turquoise	−0.26	−0.14	0.27	0.01 (n.s.)
green	−0.24	−0.02 (n.s.)	0.33	−0.17
pink	0.17	0.05 (n.s.)	−0.22	0.07
black	−0.15	−0.13	0.43	−0.30
magenta	−0.13	−0.11	0.07	0.12
purple	0.08	0.13	−0.25	0.16

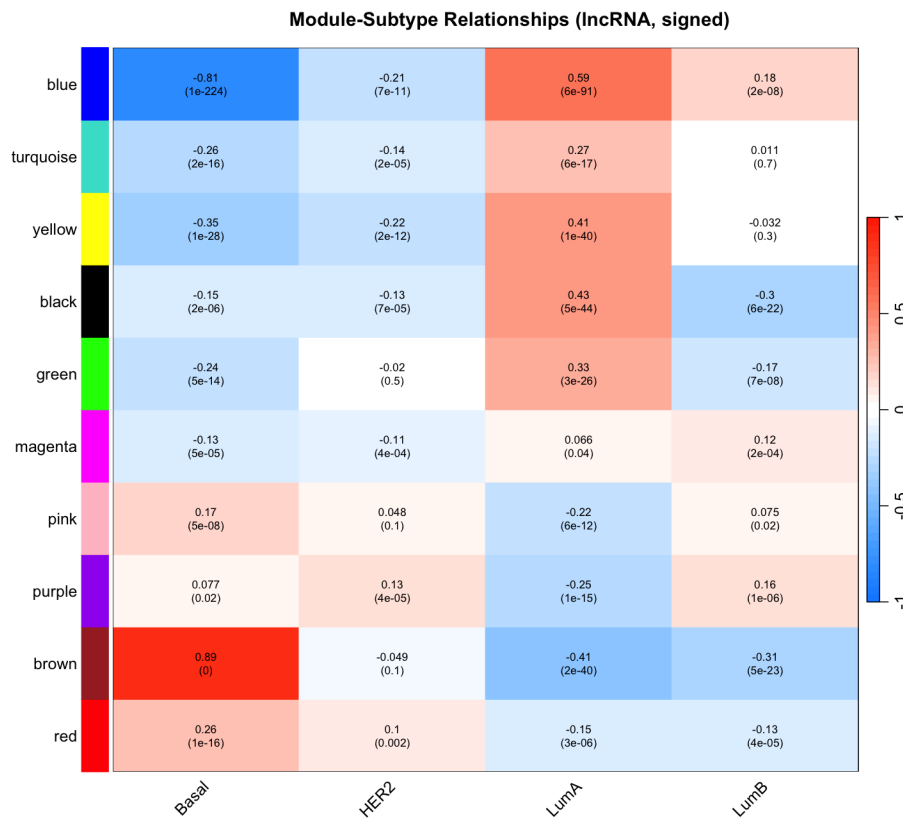


Figure 8: Module–subtype relationships, signed lncRNA network. Each cell reports Pearson r (top) and p -value (bottom).

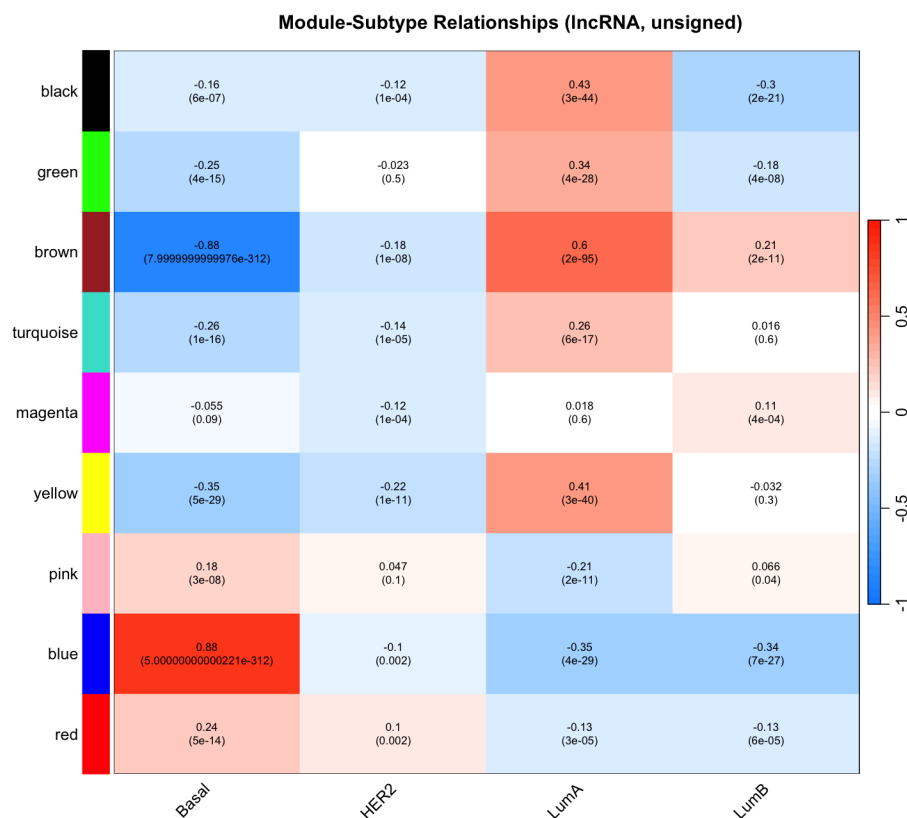


Figure 9: Module-subtype relationships, unsigned lncRNA network. Each cell reports Pearson r (top) and p -value (bottom).

2.8 Hub Genes

The top kME hub gene per signed module was: brown – *FOXCUT* (kME 0.880); blue – *LINC00504* (0.861); yellow – *RAD51-AS1* (0.854); turquoise – *NPTN-IT1* (0.851); pink – *AL121832.2* (0.851); red – *LINC00996* (0.841); green – *MAGI2-AS3* (0.834); black – *LINC02884* (0.825); magenta – *STPG3-AS1* (0.808); purple – *LINC02820* (0.786). No gene appeared in more than one module in either network.